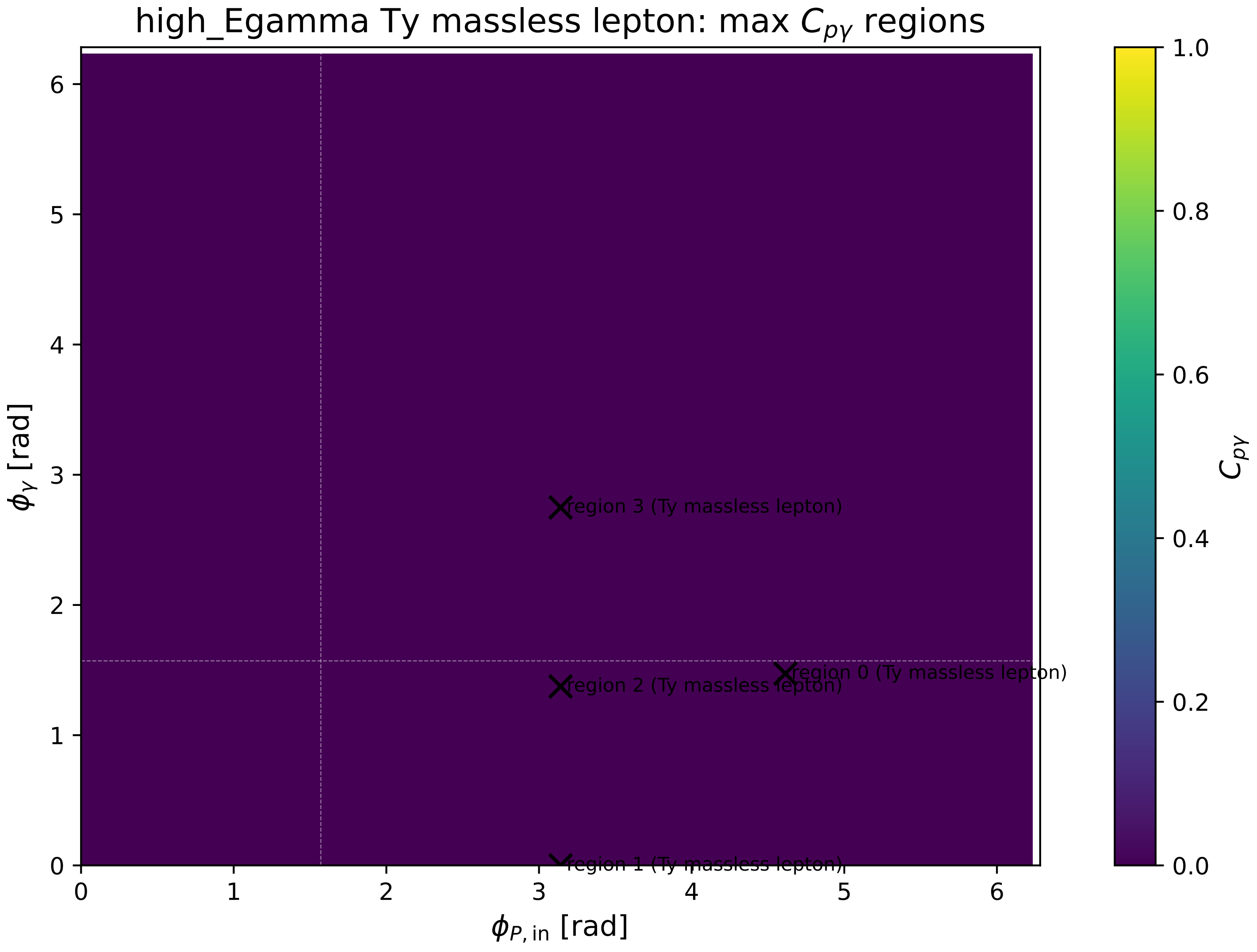
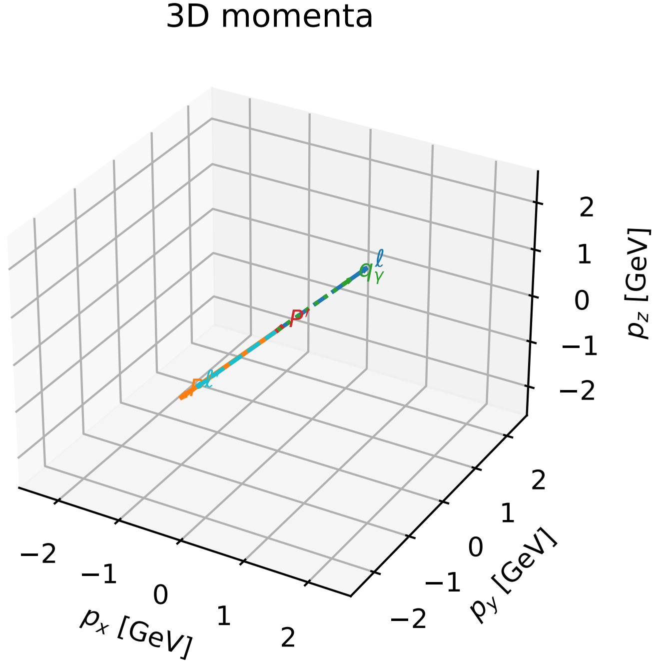


high_Egamma Ty massless lepton: max $C_{p\gamma}$ regions
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
region 3 (Ty massless lepton)
region 0 (Ty massless lepton)
region 2 (Ty massless lepton)
region 1 (Ty massless lepton)
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

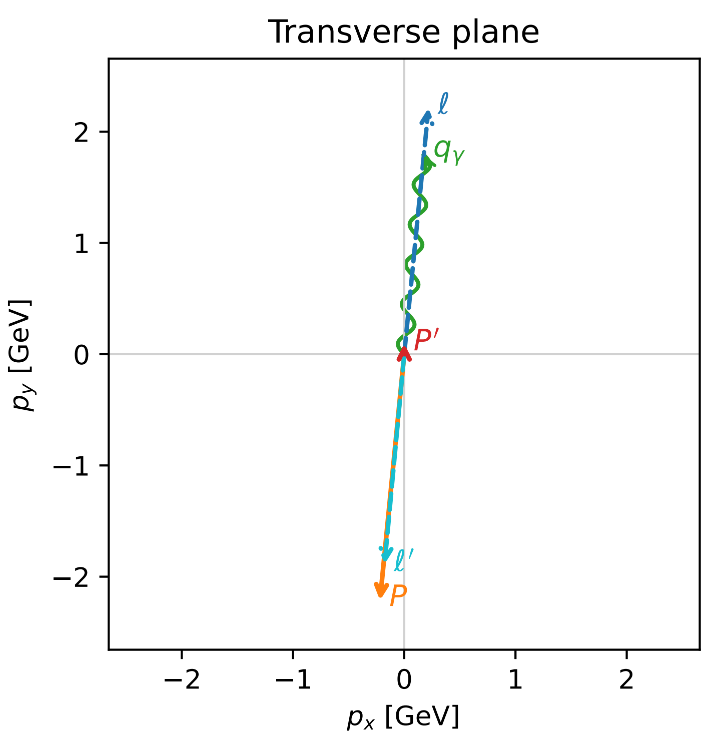
Ty massless lepton characteristic max $C_{p\gamma}$ configuration

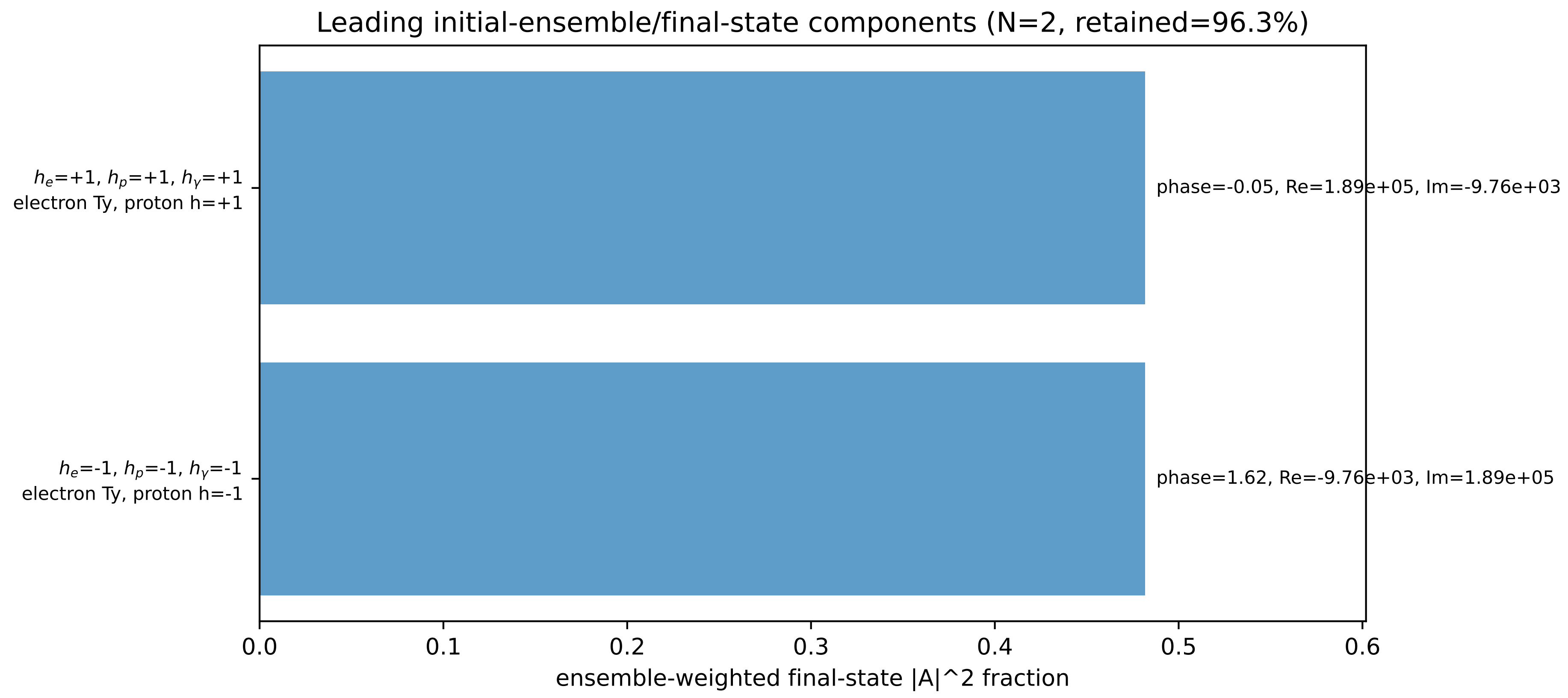


c_p_gamma_high_egamma_ty_lepton_region_0 C_{p\gamma}=3.83804e-13
region: high_Egamma_Ty_lepton_region_0 final-pair delta_xy=0.0981748 rad
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0}, h_p)$

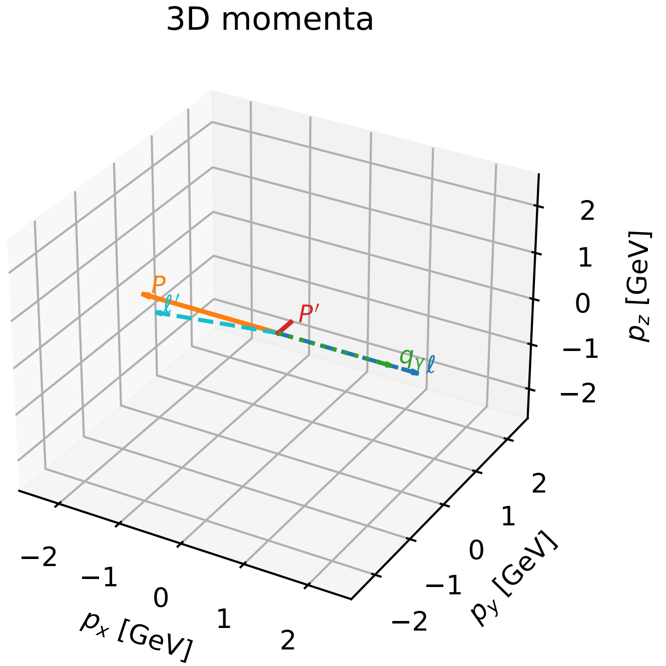
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0960515
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=1.8, ϕ_γ=1.47262
Q^2=16.8664, x_B=0.846846, t=-3.21811, W^2=3.93018, y=0.965147
$\theta(\ell', \gamma)$=179.715 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.2180 p_y= 2.2137 p_z= -0.0000
P E= 2.4141 p_x= -0.2180 p_y= -2.2137 p_z= 0.0000
ℓ' E= 1.8956 p_x= -0.1764 p_y= -1.8874 p_z= 0.0000
P' E= 0.9429 p_x= 0.0000 p_y= 0.0961 p_z= 0.0000
q_γ E= 1.8000 p_x= 0.1764 p_y= 1.7913 p_z= 0.0000





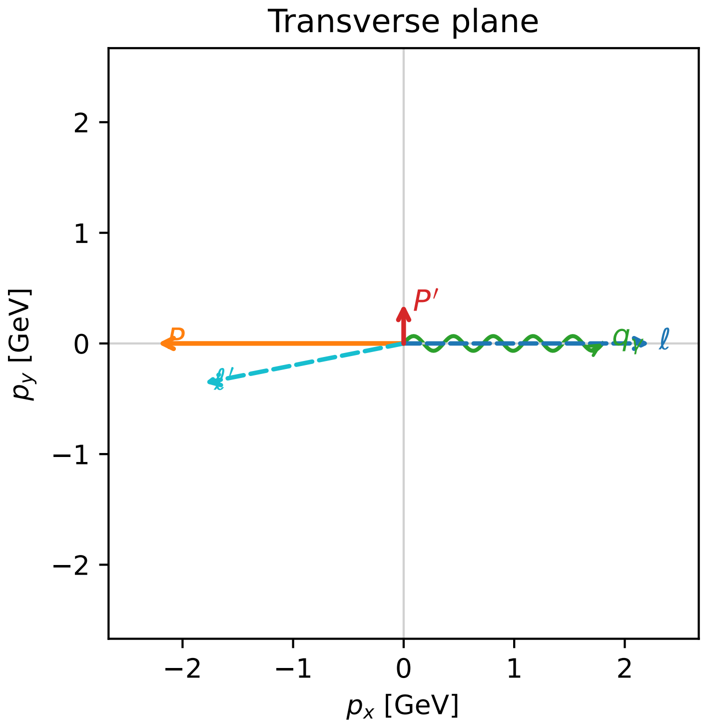
Ty massless lepton characteristic max $C_{p\gamma}$ configuration

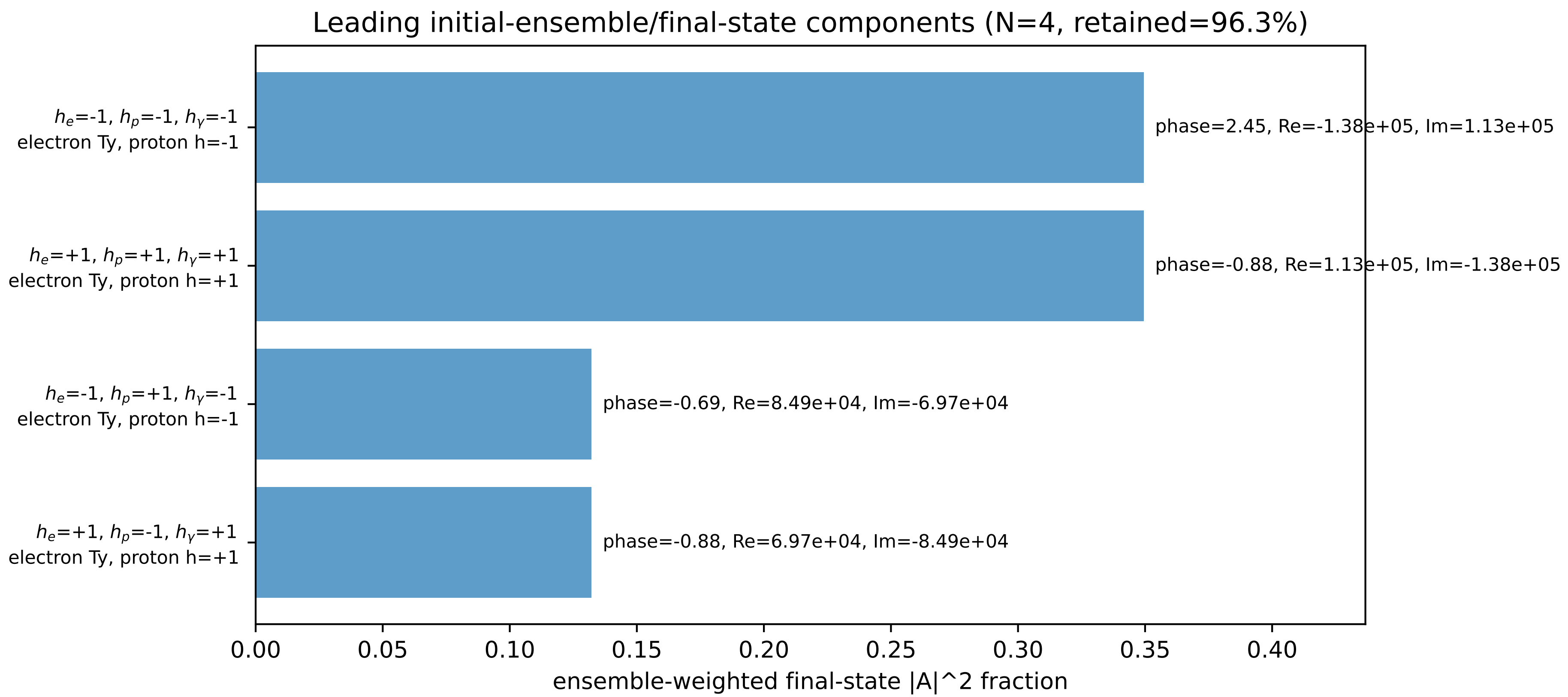


c_p_gamma_high_egamma_ty_lepton_region_1 C_{p\gamma}=0
region: high_Egamma_Ty_lepton_region_1 final-pair delta_xy=1.5708 rad
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0}, h_p)$

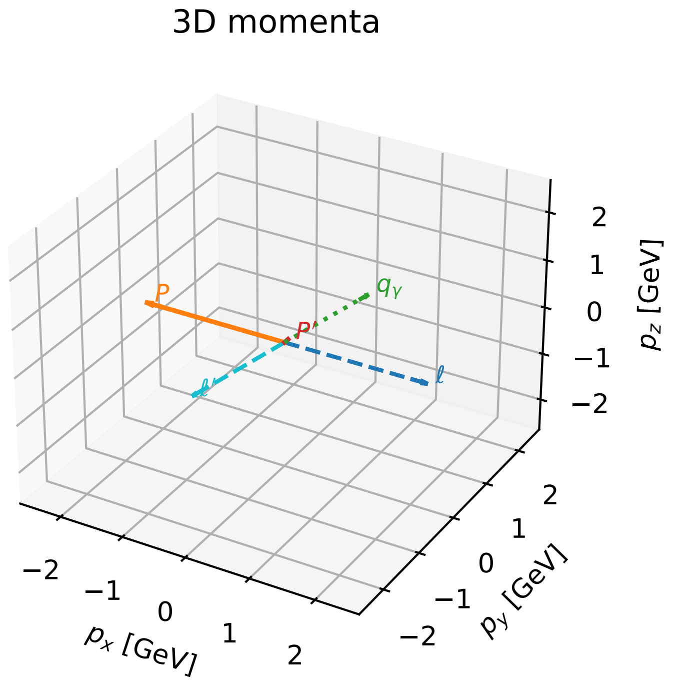
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.356666
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=0
Q^2=16.1715, x_B=0.817396, t=-3.08551, W^2=4.49252, y=0.958721
$\theta(\ell',\gamma)$=168.792 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8350 p_x= -1.8000 p_y= -0.3567 p_z= 0.0000
P' E= 1.0035 p_x= 0.0000 p_y= 0.3567 p_z= 0.0000
q_γ E= 1.8000 p_x= 1.8000 p_y= 0.0000 p_z= 0.0000





Ty massless lepton characteristic max $C_{p\gamma}$ configuration

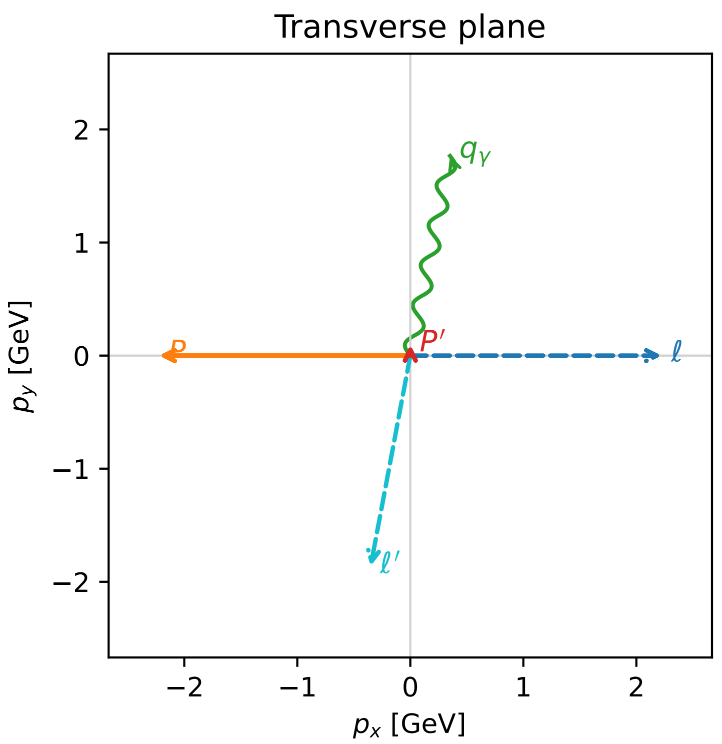


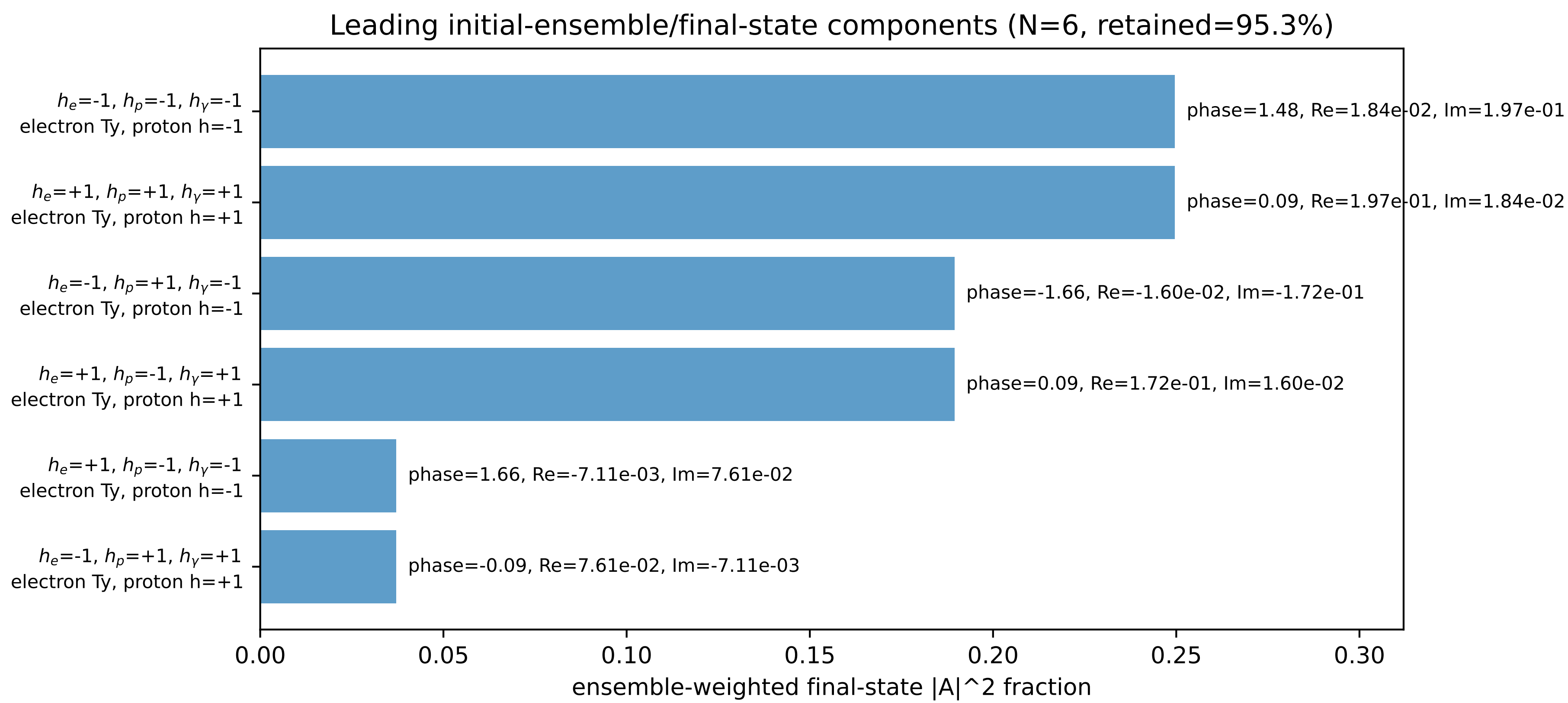
c_p_gamma_high_egamma_ty_lepton_region_2 C_{p\gamma}=0
region: high_Egamma_Ty_lepton_region_2 final-pair delta_xy=0.19635 rad
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0}, h_p)$

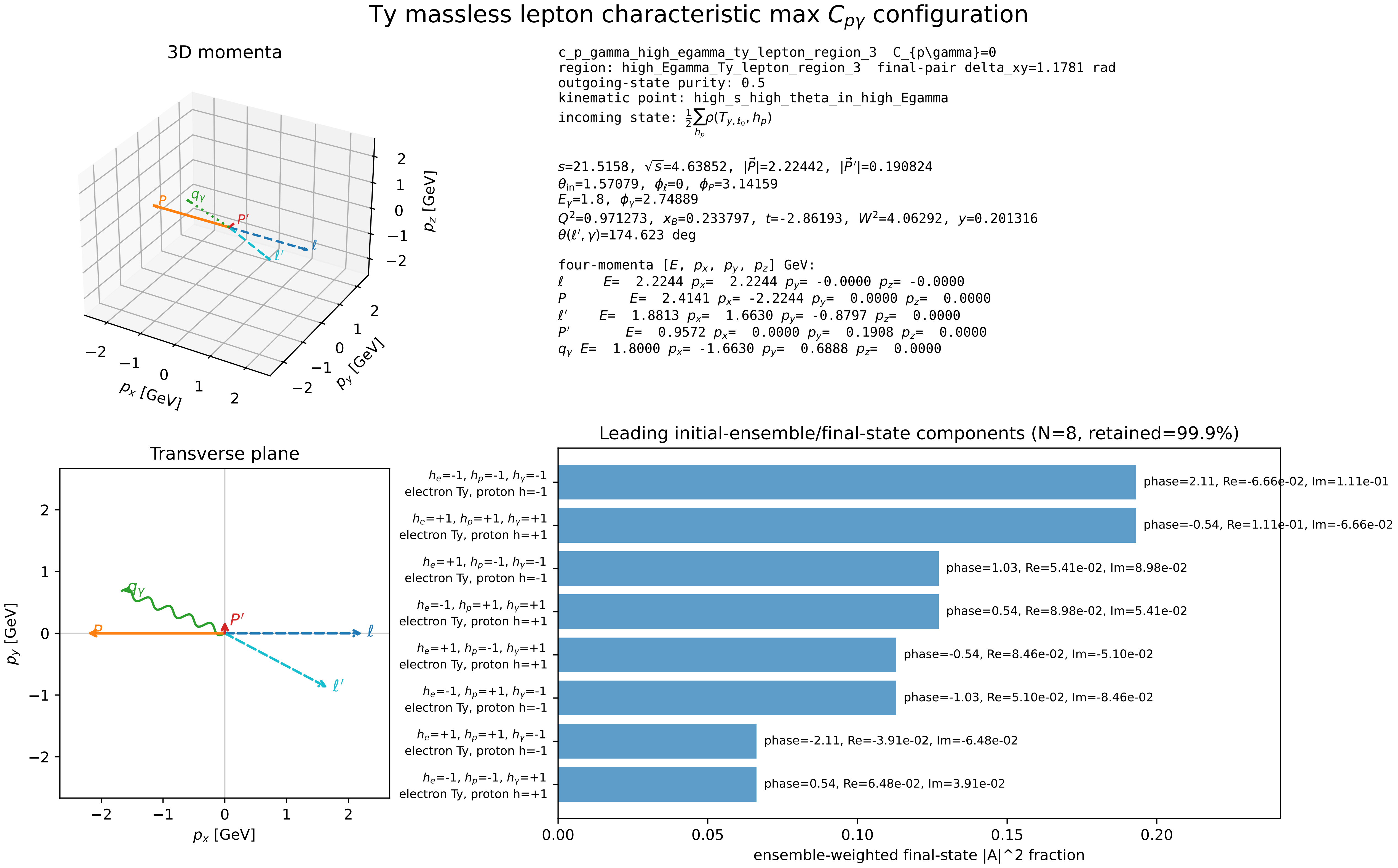
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.0972622
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=1.37445
Q^2=9.99498, x_B=0.766106, t=-2.79344, W^2=3.93133, y=0.632219
$\theta(\ell', \gamma)$=179.426 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= 2.2244	p_y= -0.0000	p_z= -0.0000
P	E= 2.4141	p_x= -2.2244	p_y= 0.0000	p_z= 0.0000
ℓ'	E= 1.8955	p_x= -0.3512	p_y= -1.8627	p_z= 0.0000
P'	E= 0.9430	p_x= 0.0000	p_y= 0.0973	p_z= 0.0000
q_γ	E= 1.8000	p_x= 0.3512	p_y= 1.7654	p_z= 0.0000






Ty massless lepton characteristic max $C_{p\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_p_gamma_high_egamma_ty_lepton_region_3 C_{p\gamma}=0
region: high_Egamma_Ty_lepton_region_3 final-pair delta_xy=1.1781 rad
outgoing-state purity: 0.5
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0}, h_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=0.190824
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=1.8, ϕ_γ=2.74889
Q^2=0.971273, x_B=0.233797, t=-2.86193, W^2=4.06292, y=0.201316
$\theta(\ell', \gamma)$=174.623 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.2244 p_y= -0.0000 p_z= -0.0000
P E= 2.4141 p_x= -2.2244 p_y= 0.0000 p_z= 0.0000
ℓ' E= 1.8813 p_x= 1.6630 p_y= -0.8797 p_z= 0.0000
P' E= 0.9572 p_x= 0.0000 p_y= 0.1908 p_z= 0.0000
q_γ E= 1.8000 p_x= -1.6630 p_y= 0.6888 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=8, retained=99.9%)
h_e=-1, h_p=-1, h_γ=-1 electron Ty, proton h=-1 phase=2.11, Re=-6.66e-02, Im=1.11e-01
h_e=+1, h_p=+1, h_γ=+1 electron Ty, proton h=+1 phase=-0.54, Re=1.11e-01, Im=-6.66e-02
h_e=+1, h_p=-1, h_γ=-1 electron Ty, proton h=-1 phase=1.03, Re=5.41e-02, Im=8.98e-02
h_e=-1, h_p=+1, h_γ=+1 electron Ty, proton h=+1 phase=0.54, Re=8.98e-02, Im=5.41e-02
h_e=+1, h_p=-1, h_γ=+1 electron Ty, proton h=+1 phase=-0.54, Re=8.46e-02, Im=-5.10e-02
h_e=-1, h_p=+1, h_γ=-1 electron Ty, proton h=-1 phase=-1.03, Re=5.10e-02, Im=-8.46e-02
h_e=+1, h_p=+1, h_γ=-1 electron Ty, proton h=-1 phase=-2.11, Re=-3.91e-02, Im=-6.48e-02
h_e=-1, h_p=-1, h_γ=+1 electron Ty, proton h=+1 phase=0.54, Re=6.48e-02, Im=3.91e-02
ensemble-weighted final-state |A|^2 fraction